

A THEORY OF FINITE REPRESENTATION

Part II

Finite Reports, Accounts and
Operational Insufficiency

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Geofinitism and Finite Symbolic Mechanics

Draft note

This second part follows *A Theory of Finite Representation, Part I: Extent, Registration and the Licence to Represent*. Part I described the passage from interaction to finite registration, the development of Functional Symbolic Trajectories, and the licences by which compressed symbols become available for further use. The present volume asks how such trajectories are assembled into accounts, how a compact value can travel without its account, and when the missing distinctions become consequential for the next operation. Its central examples are deliberately elementary. The reader need not accept a proposed physics of light in order to inspect the fate of a half in subtraction, a quarter in multiplication, or a ledger that supports one further cut but not another.

This is a working construction. The finite account, conditional insufficiency witness and debt register are offered as provisional instruments of thought. Examples use a stipulated bead resolution; they do not assert a universal particle size or an edge of zero width. Classical equalities retain their exact meaning within the classical symbolic basin. A claim about a measured registration is related to that basin through declared conditions of correspondence, accessible history and uncertainty. In particular, the physical application considered near the end of this draft remains an audit of how a spectral comparison becomes reportable, not a completed alternative law of spectral change.

The style of this volume is intentionally sequential. A notation is introduced only after the activity it abbreviates has been described. The equations are parts of a finite written report, and the report itself has an author, a carrier and a history. The appendix entitled *Prior explorations, September 2026* preserves the route through which this draft was formed; it is not an errata list or a substitute for the arguments in the chapters.

Abstract

An arithmetic result can be correct under its rules and insufficient as an account of a finite action. Subtracting one half from a whole yields one half as a value, but that remainder does not name the disposition of the part removed. Multiplying two halves yields one quarter, but distinct finite cutting ledgers can yield that same value and support different subsequent cuts. This volume develops these examples into a theory of finite reports. A finite report is a bounded, instantiated formal document that assembles one or more Functional Symbolic Trajectories into a coherent structure for a declared purpose. A reported value is a smaller symbolic result extracted from such a structure. The distinction prevents the word *report* from being compressed into one numeral.

The proposed account distinguishes the bead path, which follows admitted finite distinctions and actions; the ledger path, which keeps their order, disposition and provenance; and the classical path, which operates on reported values. Operational insufficiency is witnessed when two admitted finite ledgers yield the same value but a named next action is admissible for only one, or their continuations fail a declared correspondence test. Representational debt names the specific omitted or unavailable distinctions needed by a later claim. Neither insufficiency nor debt is assumed to be an absolute property of a number. The volume follows addition, subtraction, multiplication, division, mathematical proofs and scientific reports, then treats spectral comparison as an opening bridge case. It closes at the threshold of a further study of FSM mathematical objects.

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Notation and reading conventions

Part I used α for an operative distinction condition, δ for uncertainty, H for accessible history and C for admissibility and licence. These meanings remain. A single α need not be assigned to an entire document: distinct registrations within it may have distinct conditions. The interactional bar $A |_C B$ marks a conditioned correspondence between registrations or between a model and a registration. The ordinary equals sign remains in classical arithmetic and in definitions internal to a declared symbolic system; its use there does not assert identity of the histories of two measured states.

A ledger L is an accessible finite record of actions and their relevant registered distinctions. The notation $\text{val}_C(L) = v$ means that a value v is extracted from that ledger under a reporting rule C . The numeral v is a classical value within the rule, carried on a finite written surface. A *finite report*, written $\mathcal{F}$, is a coherent formal document composed of one or more FSTs, their admitted relations and their stated scope; it is not synonymous with v . A proposed ledger continuation is written $T_{a,C}(L)$, where a names an action. This expression may be undefined when the action cannot be performed under C . An apparent inverse of a compressive operation will always require its own conditions and available record.

Prologue: The other half

We begin with a line that no school exercise needs to dispute:

$$1 - \frac{1}{2} = \frac{1}{2}. \tag{1}$$

The line is short because the arithmetic task is short. Under its rules it reduces the expression on the left to the value on the right. If the question concerns only that value, the work is done. But a second question can be put to the same sentence. Suppose the one names a particular registered whole and the subtraction names an act of taking. Which half remained, which half was set aside, and what record permits either to be identified again? The right side of Equation (1) does not answer. The question did not make the arithmetic false; it changed what the sentence was being asked to account for.

The distinction is familiar whenever a transfer has consequences beyond its final total. If one unit of gold is registered in a vault and a half-unit is removed, an account of the vault may quite properly report a half remaining. An account of the transfer must additionally hold the party, destination, division procedure, uncertainty and any residue. That further account can itself be incomplete. It will hold only those events the relevant registration and record made available. Nevertheless, it is longer than the numerical remainder and can support questions the remainder cannot. The earlier working document *Finite Accounts and the Discarded Half* used this specimen to make the second column visible [2].

The present inquiry follows the difference between the right answer and the adequate account. It starts with beads because a finite sequence of chosen marks can be followed without first granting an invisible continuous line or a ready-made object behind the word *number*. It proceeds to reports because mathematical and scientific claims rarely travel as solitary numerals. They travel in articles, proofs, files, notebooks and books, each assembling multiple symbolic trajectories into a structure a reader can reconstruct. The question *Where has the other half gone?* becomes, in that wider setting, a question about which distinctions a report carries and which it leaves to a later handler.

Chapter 1

From finite representation to finite reports

1.1 The middle established in Part I

Part I described a route from an interactional trajectory to an instrumental change, registration, symbolic transformation, model and scientific statement [1]. That route was not intended as a succession of containers through which an unchanged object passes. Each crossing selects a distinguishable relation under a finite window. A recorded value is therefore already a late position in a trajectory. It has become available only after decisions about what counts, what may be compared and what can be carried forward. The three basins of Part I keep the interactional registrations, endogenous symbolic constructions and their bridge rules related without declaring them identical.

The Functional Symbolic Trajectory, or FST, carries this account through language as well as mathematics. A word, diagram, line of a proof or instrument code can constrain what happens next. Compression permits one bead to stand for a longer chain; decompression permits a current handler to build a further chain from that bead. Neither operation guarantees that the entire previous history is preserved or recovered. Part I treated this as a condition of symbolic mobility. Part II asks what happens when several FSTs have to support a decision together, especially when one of their values is detached and sent onward alone.

1.2 The finite report as a formal document

We shall call a *finite report* a bounded, instantiated formal document that gathers one or more FSTs into a coherent structure for a declared purpose. The word *document* is used widely. A proof on a page, a laboratory notebook, a file of registered instrument states, an annotated image, a table with its calibration record, or a carefully composed verbal account may each serve. A report has a carrier and a scope. It includes selected relations among its trajectories, a way of reading them, and conditions under which its conclusions may travel. Its coherence is an achievement within those conditions; it is not a claim that every contributing event has been written down.

One may schematically write

$$\mathcal{F}_C = (\Gamma_1, \dots, \Gamma_m; J, H, C), \quad m < \infty, \quad (1.1)$$

where each Γ_i is an accessible finite trajectory, J records the admitted connections among the trajectories, H records accessible provenance and C identifies the report's licence and scope. Uncertainty and distinction conditions belong at the registrations and transformations to which they apply; they need not be falsely replaced by a universal error bar around the document. Equation (1.1) is a reading aid, not a claim that one tuple exhausts every possible form of a report. The document before the reader is itself a finite report in this provisional sense.

1.3 Reported value and declared purpose

A reported value is a particular extraction from a ledger or report. It is a smaller object than the document that supports it. The value $\frac{1}{4}$ may be a true answer under a fractional rule while saying little about the beads retained in the particular construction. A spectral parameter may be a valid summary of an admitted comparison while being too sparse for a new question about calibration history. It is therefore misleading to use *report* and *value* interchangeably. In what follows $\text{val}_C(L) = v$ names a value extraction under the conditions C ; $\mathcal{F}_C$ names the formal document that can carry this and other related trajectories.

The phrase *finite report* may sound modest, but it bears considerable work. Reports can be copied, abridged, corrected, joined and cited. Each operation changes which parts of their FSTs a later reader can inspect. A report sufficient for a classroom sum may be insufficient for an audit of material transfer; a proof sufficient for an internal theorem may be insufficient for a physical application of its parameters. No final universal report is assumed. A report closes only with respect to a stated task and remains an available bead for further inquiry.

Chapter 2

Beads, ledgers and the classical path

2.1 Three paths through one calculation

A bead in this work is a finite symbolic distinction capable of occupying a position in an accessible chain, carrying some provenance and conditioning a later operation. It is not declared an ultimate material atom. To say that a bead cannot be divided is to state a resolution and a licence for a particular construction. A new registration may later distinguish what an earlier act could not, but that new act must be entered in the account. This prevents a convenient drawing of a circle or point from silently becoming a claim of infinitely sharp boundaries.

There are three paths to follow. The bead path attends to the particular registered distinctions presented and the acts performed on them. The ledger path records their order, what was retained, what was moved aside and which distinctions remained available. The classical path manipulates values under a specified arithmetic rule. The paths can correspond. They need not be the same trajectory. A traditional computation often retains the third path because it is sufficient for its immediate purpose; a finite report can keep the other two available when a subsequent purpose requires them.

For a bounded exercise, let L_n denote the accessible ledger after n admitted actions. It may be read as the present bead arrangement together with an ordered event record and its available conditions of registration. We write

$$L_n \xrightarrow{T_{a,C}} L_{n+1}, \quad \text{val}_C(L_n) = v_n, \quad (2.1)$$

where $T_{a,C}$ may fail to be admitted for a particular L_n . By contrast, an endogenous arithmetic operation f_a may take the reported value to $f_a(v_n)$ whenever its formal domain allows. The relation between $\text{val}_C(L_{n+1})$ and $f_a(v_n)$ is a bridge question. It cannot be settled by replacing the ledger with the value before the action has been examined.

2.2 A ledger is selective

The ledger must not be imagined as a complete transcription of an unreachable world behind the symbols. It is itself a finite symbolic construction. An observer may not record the initial formation of a mark, an instrument may register only within a selected window, and a later editor may preserve a summary rather than every intermediate state. The

account nevertheless improves when it states what it can and cannot follow. A set-aside portion already recorded in another column has a different status from a portion whose numerical magnitude can be inferred, and both differ from an event that never entered any accessible registration.

This selectivity also prevents the ledger from becoming a moral injunction to write everything down. A useful account depends upon the next question. A narrow calculation may require only a value and rule; a claim about an instrument may require calibration and excluded registrations; a claim about where gold went requires a contra-entry. The ledger's function is to make the basis of a continuation inspectable. Its length is a cost as well as a resource, and the report must say which cost it has chosen to carry.

Chapter 3

Addition and the first compression

3.1 Joining finite chains

Addition provides a first, unusually clear constructive story. Imagine two chains of registered beads, one with two admitted members and one with three. They may be joined, counted under a common rule and reported as five. The classical equality $2 + 3 = 5$ does not require the particular beads to retain names. Its value is reproducible because the counting licence identifies many concrete arrangements as instances of the same sum. That identification is an extraordinary instrument: one no longer has to reconstruct each history before using the value five in a later proof.

Still, the operation of joining is more than the terminal numeral. The chains can differ in provenance, their beads can have different admissible properties, and the order of joining may matter to a later action even when it does not affect the count. The numeral five names a value under a counting convention. If a later question asks which three beads came from the second chain, the report must have retained that division or the question will require a new registration. This is a first form of conditional insufficiency, arising before fractions or subtraction make the set-aside portion especially vivid.

The same visible sum can also be written $1 + 4 = 5$. One can therefore follow two distinct ledger trajectories to the same reported value without claiming any difficulty in the equality of the results. The operation $+$ compresses assembly under its rules. An account of the particular assembly can remain available, disappear or be reconstructed only incompletely through a later handler. It is important not to mistake the number of glyphs in the equation for the number of beads in the joined collections. Inscriptional beads, counted beads and measurement registrations are three related but differently licensed inventories.

3.2 Repetition and its boundary

Repeated addition permits a familiar construction of whole-number multiplication. To form $3 \times \frac{1}{2}$ one may assemble three admitted halves under a stable reference whole and report $\frac{3}{2}$. This story depends upon an integer count of repetitions. When a multiplier is itself a fraction, the phrase *repeat the addition half a time* does not describe the same admitted procedure. Classical arithmetic extends multiplication consistently across its numerical domain; a finite account asks what action a particular application of that extended symbol represents.

That transition matters for the following chapters. Multiplication may assemble copies, select a proportion or compose scaling rules. The numerical sign stays stable while the constructive route changes. Addition has not failed; a wider operation has been licensed and the visible formula does not by itself reveal which route supplied its meaning. The value remains useful, and its specific ledger may become relevant when the result is asked to support a further action.

Chapter 4

Subtraction and the contra-entry

4.1 A correct remainder

Return to $1 - \frac{1}{2} = \frac{1}{2}$. The left side shows the whole and the removed amount, so an editor who retains the entire expression has not erased every clue about the numerical event. If only the right side is carried forward, however, many different starting values and actions could have produced it. Even with the original expression in view, the calculation does not name a particular half, its destination, the grain of the cut or the record by which the halves could be distinguished. These are not objections to subtraction as a formal operation. They are features of the task for which subtraction is being used.

An account of a particular partition can initially be written as

$$W \mapsto (R, X; h), \quad \text{val}_C(W) = \text{val}_C(R) + \text{val}_C(X), \quad (4.1)$$

where W is the registered whole, R the retained portion, X the portion set aside and h the accessible event record. The equality on the right is a value relation under C ; it does not identify the histories or carriers of W , R and X . If the operation involves a kerf, loss in handling or an uncertain comparison, the account must add those distinctions rather than forcing Equation (4.1) to carry more than its licence supplies. One can use the classical value as a mobility engine while retaining the account alongside it.

4.2 Where the debt begins

It is tempting to say that the right-hand side has *destroyed* the removed half. That is too quick. The portion may remain registered in the ledger, may still be present in another place, or may be unidentifiable from the value alone. The debt arises when a later claim needs an entry that the travelling result does not contain. If a vault report says only that half the gold remains, it is insufficient for a question about custody of the other half. The amount may be inferable from the initial total, but the custodian and route are not. The later claim owes an account of those distinctions.

This is the point at which the older double-entry image is productive. The second column is not a metaphysical duplicate of the first. It is a place in the finite document where a transfer can be recorded. An account that names an untracked surround still does more than claim that a removed amount ceased to matter. If the surround is not inspected, it should not be given the false precision of a known destination. The ledger must preserve both a named lack and a known entry without silently treating them as identical.

Chapter 5

Multiplication, cutting and scaling

5.1 A fraction as an instruction

A fraction can be read as a value and, under a declared whole, as an instruction for an action. The instruction $\frac{1}{2}$ says to partition an admitted extent and retain a portion corresponding to one of two equal shares. Whether such a partition is available depends upon the registered arrangement and the rule of equal extent being applied. The arithmetic value $\frac{1}{2}$ by itself contains no record of which share was selected. With two successive halvings the familiar equation

$$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4} \tag{5.1}$$

can therefore be read as a compression of two actions into one reported proportion. Its classical validity is unaffected by the further question of what was retained or set aside in a particular enactment.

The contraction is often surprising to someone taught that multiplication is repeated addition and therefore makes more. That teaching describes one route through a restricted domain. Scaling describes the common value relation: a positive factor above one expands a positive magnitude, a factor of one leaves it unchanged, and a factor strictly between zero and one contracts it. The arithmetic scale $S_r(v) = rv$ is defined within its formal domain. A finite cutting operation $T_{r,C}(L)$ has a narrower domain and an event record. The same written r can thus occupy the role of value, ratio or instruction, and its role should be declared whenever the distinction matters.

5.2 Two ledgers at one quarter

Consider a stipulated whole of eight extent units made from four unit beads and two beads of extent two. These beads are chosen symbolic registrations under a rule that permits selection of whole beads and does not permit dividing one of them during the exercise. Their stated extents are descriptors of this construction, not claims about universal atoms. The first ledger retains the four unit beads at the first halving and two unit beads at the second. The second retains the two double-unit beads at the first halving and one double-unit bead at the second. Each reports two of the original eight extent units, namely a quarter. Their accessible trajectories can be displayed without collapsing their bead

identities:

$$\begin{aligned}
L_A : (1, 1, 1, 1, 2, 2) &\longrightarrow (1, 1, 1, 1) \longrightarrow (1, 1), \\
L_B : (1, 1, 1, 1, 2, 2) &\longrightarrow (2, 2) \longrightarrow (2), \\
\text{val}_C(L_A) = \text{val}_C(L_B) &= \frac{2}{8} = \frac{1}{4}.
\end{aligned} \tag{5.2}$$

The full ledger also records that L_A set aside $(2, 2)$ and then $(1, 1)$, whereas L_B set aside $(1, 1, 1, 1)$ and then (2) . Those set-aside chains carry equal reported extents in their respective stages but different bead compositions.

Now ask for a further halving under the same rule. In L_A one unit bead can be retained and the other set aside. In L_B the retained state consists of one two-unit bead, which this rule does not admit dividing. The classical operation $\frac{1}{4} \times \frac{1}{2} = \frac{1}{8}$ proceeds. Its finite enactment proceeds on L_A and halts on L_B under the stipulated licence. A new operation could register a distinction inside the two-unit bead, but doing so would alter the available bead chain and add an event to the ledger. It cannot be silently assumed to have occurred in the earlier report.

5.3 What the specimen establishes

The example establishes a local result, not a universal prohibition on fraction multiplication. The reported quarter is sufficient for comparisons that depend only on its numerical proportion. It is insufficient to decide the admissibility of the next cut under this finite rule. The location of the insufficiency is the identification of two particular ledgers at one value, followed by a continuation that requires a distinction the value did not carry. The story also displays why a later dilation of the retained extent need not restore the original arrangement. Numerical inversion and reconstruction of the earlier chain are different operations.

Chapter 6

Division, decimal writing and the chosen unit

6.1 Several routes through division

Division can partition a whole into equal portions, ask how many portions of a given size can be formed, calculate a ratio or produce a quotient and remainder. An algorithm may use repeated subtraction, but it would be a mistake to say that every division is literally the same finite taking. The symbol $\div$ gathers different permitted procedures in the classical basin, while the report of a particular procedure may need to state its reference whole, resolution and remainder. Our interest lies in the passage from these actions to one portable value, not in replacing each familiar algorithm with one mandatory story.

The decimal 0.5 makes a half available through a base-ten inscription. One may write $5/10$ and, under a permitted reduction, $1/2$. The inscription shows an internal place-value construction but does not identify a unique cutting history. A decimal expansion may be longer than the fraction that names the same rational value. A change of numeral base changes the written bead chain and the cost of carrying it; it does not, by mere re-expression, change the value of the ratio. If the change also alters a finite encoding, rounding or registration operation, those further acts must be placed in the ledger rather than attributed to the base alone.

6.2 Division at a finite resolution

The two-unit bead in the earlier specimen makes the issue concrete. Classical division by two yields one extent unit. The stipulated operation on a single registered bead has no admitted half-bead state. We should say that the proposed cut is *insufficiently supported at this resolution*, not that one is an impossible number. A finer instrument or a revised symbolic licence could produce new beads; the event of refinement would then be part of the new account. The earlier ledger cannot acquire it retrospectively through arithmetic notation.

A measured number must consequently be more than an exact real value with an ornamental error sign attached. Its reference, finite inscription, available distinction, uncertainty, provenance and admitted use belong to its account. Part III may propose a fuller notation for such an FSM object. This volume leaves the object open and asks a narrower question: when the value of one finite registration is used in a subsequent

operation, what previous distinctions does that operation actually require?

Chapter 7

Conditional sufficiency

7.1 The finite witness

The expression *information loss* is too general to settle the question we have reached. A two-character value may discard the shape of a long chain, yet support every operation presently proposed of it. Conversely, an omitted distinction that looks minor in the first exercise may govern the next one. We therefore define sufficiency relative to a continuation. Let a be a particular proposed action and C the conditions under which a ledger and its reported value are being used. Two exhibited ledgers L_A and L_B constitute an *operational insufficiency witness* for the extracted value v if

$$\text{val}_C(L_A) = \text{val}_C(L_B) = v, \quad \text{Adm}_C(T_{a,C}, L_A) \neq \text{Adm}_C(T_{a,C}, L_B), \quad (7.1)$$

where Adm_C is the declared admissibility judgement for the next action. A second form of witness occurs if both continuations are admitted but their resulting registrations fail a correspondence required for the particular claim. This second form must state the comparison and its tolerance. It cannot be established merely by pointing out that the ledgers have different stories.

Equation (7.1) makes a limited and useful assertion. It takes two particular finite records, one value rule and one proposed continuation. It does not quantify over an imagined completed collection of all possible histories. It says that a user who sees only v cannot determine the admitted outcome of a for those two cases. The quarter-bead story witnesses precisely this situation. The same value can still be sufficient for calculating another classical proportion under its endogenous rules. Insufficiency belongs to a stated operation and purpose, not to the glyph $\frac{1}{4}$ in isolation.

7.2 The other direction

It is also useful to ask when an extracted value may safely be carried alone. Suppose a report declares a bounded family of admitted ledgers and a particular class of continuations. If every ledger in that examined family supports the relevant continuation and the resulting extracted values meet the declared correspondence test, the compact value is sufficient within that stated scope. The scope is part of the result. A new instrument, finer distinction or differently worded task may reopen it. A report that omits this scope can make a local achievement appear to be an unlimited licence.

This two-sided formulation protects the arithmetic from a false accusation. Classical

multiplication of rationals is internally consistent and can continue indefinitely within its formal construction. A bead exercise under finite cutting rules has a different admitted domain. When the two are brought into relation, the bridge must say what it preserves. An exact answer in one basin and a halted action in another are not a contradiction until a claim has asserted that the basins are identical in precisely that respect.

7.3 A theorem-shaped question

The principle invites a more formal programme. For a declared family of finite ledgers, can one identify the smallest accessible record needed to decide a named continuation? How stable is that decision under a change of resolution, handler or reference whole? A general theorem cannot be claimed from one specimen. But a finite witness can defeat an overbroad sufficiency claim, and a bounded audit can establish a local one. The task of Part II is to make these proof obligations speakable before claiming an exhaustive calculus of FSM objects.

Chapter 8

Representational debt and recoverability

8.1 When an account becomes owing

Debt in this monograph is an accounting relation, not an extra substance located behind the symbols. An extracted value incurs no moral or physical penalty by being short. A debt becomes relevant when a subsequent claim relies on a distinction omitted from the carrier presented to support it. If an answer about a retained half is used to assert the destination of the other half, the claim owes a destination record. If a spectral comparison is used to assert that two line identifications have one common history, the claim owes the correspondence and calibration conditions needed for that assertion. The debt is indexed to the use, not stamped forever upon the number.

A debt entry should name the distinction required, the stage at which it was last accessible, its present status, and the proposed action or claim needing it. This is an ordered ledger of obligations. Some entries concern a set-aside bead that is still available in another column. Some concern a provenance field present in an earlier file but absent from the figure being cited. Some concern a registration never made, and cannot be discharged by a more confident restatement of the final value. Treating these cases as one scalar *information deficit* would hide their different possible remedies.

8.2 Uncertainty of value and uncertainty of history

Two quarters can agree exactly under a classical valuation and remain different as accessible chains. Thus numerical value uncertainty and uncertainty about provenance need not move together. If a ledger is discarded, the set of compatible construction stories may broaden for a later reader, even while the reported numeral retains all its digits. Later independent constraints can narrow that uncertainty; nothing here establishes that it must increase monotonically. If the event was never registered, a plausible reconstruction may still be built through a handler, but it cannot honestly be presented as the original recovered trajectory.

The Alphonic condition α and uncertainty δ also perform different roles. An operative distinction condition says which differences were registered as non-single in an act. Uncertainty describes the limits of a particular claim formed from such acts. Neither should be offered as a universal microscopic length. A report can carry different windows and

distinction conditions at different stages, and a later comparison may choose a resolution appropriate to its task. This remains true even when a compressed output shows just one apparently exact value.

8.3 Cost, density and a deferred measure

There is an important future problem of representation cost. Longer bead chains require storage, reading, checking and perhaps interactional work. A one-line value compresses that burden. It is tempting to add the missing history into a single debt or density variable, but no common unit has yet been constructed for displaced material, omitted provenance and symbolic processing cost. Part II keeps the entries separate. A later programme may define a cost for a specific operation and carrier, compare two finite reports under that cost, and investigate how their accessible histories change. Until then, *debt* names what is owed by a particular assertion, and *insufficiency* names a witnessed failure of support.

Chapter 9

The mathematical report

9.1 A proof as a carried trajectory

Mathematical proofs are exemplary finite reports. They contain inscriptions, definitions, lemmas, substitutions and decisions about which prior results may be invoked without unfolding their entire derivations. An equality licenses one expression to occupy the operative position of another under stated rules. A lemma can stand for many lines of earlier argument. The reader's training supplies a handler through which the abbreviated proof becomes traversable. Its brevity is not an absence of mathematics; it is a selected way of carrying mathematical history.

The result at the end of a proof may be stable within the symbolic basin even when the proof takes a quite different route from another proof of the same theorem. Different routes may preserve different intermediate objects and suggest different extensions. The last statement can be sufficient for one subsequent proof and insufficient for a question about which constructions, dependencies or algorithms produced it. For that second question, the proof document and its accessible references matter. A report of a theorem is therefore not identical to the theorem regarded as a portable formal conclusion.

9.2 Cancellation and the licence to ignore

Algebraic cancellation provides a small specimen. In an expression such as

$$(a + b) - b = a, \tag{9.1}$$

the formal rule is licensed for the relevant algebraic objects. The visible b on the right is gone because the form of the answer was reduced; it has not established that a particular material b was annihilated. A physical transfer, a numerical implementation with finite precision, or a manipulation of registrations may require a separate ledger. The question is not whether Equation (9.1) ought to be forbidden. It is whether a later claim about a measured event has silently borrowed its exact cancellation without declaring the bridge.

The historical practice of moving between geometrical constructions and algebraic expressions makes this worth studying without attributing the entire change to one founder. Descartes explicitly selected a line segment as a unit and constructed multiplication of segments by proportion in *La Géométrie*; the algebraic name and the diagrammatic construction coexist in that text [3]. Wallis later used directed movement along a line to make negative quantities interpretable, bringing a journey and a terminal position into one

written example [4]. The productive question for us is what a later notation retains from such constructions, not which historical person allegedly invented the deletion of extent. A full historical treatment belongs to the future scholarly edition of this programme.

9.3 A proof crosses only by a bridge

Part I distinguished internal mathematical identity from conditioned comparison with a registration. That distinction remains when a proof reaches an application. A theorem established over ideal functions may organise the prediction of a finite measurement. Its internal success does not, by itself, specify the sensor, window, calibration, uncertainty or admissible comparison. Those are additional FSTs in the scientific report. To document them is not to diminish the proof. It is to explain the particular crossing under which the proof's result becomes effective outside its original basin.

This is why the present volume neither bans classical notation nor replaces every equals sign by a tolerance sign. The signs have different jobs. Within the proof, $=$ performs the declared formal identity. At the bridge, $|_C$ reminds us that model and registration correspond under conditions whose finite documentation matters. If a report compresses those jobs into one mark, a later reader may be led to demand an identity that the experiment never registered.

Chapter 10

The scientific report

10.1 From registration to shared document

An instrument does not hand a complete natural object to a document. An interaction changes a sensor or other registering arrangement; a finite code is assigned; corrections and comparisons are carried out; and a report permits a community to read the result. Each stage may preserve a record, discard it, or compress it into a parameter. A scientific report gathers several trajectories: the interactional registration, an instrument's own history, calibration, numerical transformations, exclusions, uncertainty and the argument that makes a new claim. Its coherence may be strong enough to support a decision while still being bounded in scope.

Consider a detector array. A displayed image can be one of several finite reports built from the same recorded array. Choice of exposure, background, scaling, interpolation and colour map can alter the visible distinctions that a later reader makes. The unmodified array, if retained, allows further constructions but is already an instrument record rather than unmediated access to a completed object. The display's legend, reduction log and uncertainty record may therefore matter to what the image licenses. The term *image* compresses these operations into a familiar noun; an audit can unfold it when a new question demands more.

10.2 The generonic chain

Part I identified the analogue-to-code transition as an important part of the generonic boundary. An ADC is particularly explicit: a range of response states is assigned to a finite alphabet, and thresholds determine which differences become separately reportable. But the ADC is not the whole boundary. The sensor before it, the sampling window, the gain, and the software transformations after it all condition the registration. A reported number can be stable even when a different part of that chain would yield a different admission decision for a later task.

The scientific report should record which version of a registered trajectory has been carried forward. A raw frame may be accessible while a paper presents a co-added, calibrated spectrum; a quality mask may still name rejected pixels; a fitting routine may have retained several candidate solutions. In such cases the difference is absent from the published summary but present elsewhere in the documented chain. When an event was below the operative distinction condition, no later multiplication of decimal places can

retroactively create its earlier registration. Both cases deserve an account, but their debts are different.

10.3 A report can earn a stronger bridge

The report need not include every original event in order to be reliable. Independent instruments, repeated measurements, standards and alternative reductions can provide overlapping paths that constrain an interpretation. The bridge becomes stronger when the same conditioned claim survives these changes and the report makes their relationships inspectable. That strength is not a single eternal number. It belongs to the range over which the comparisons have been made and to the future tests that the report can help design.

Chapter 11

A spectral comparison as an opening case

11.1 Begin with two reportable trajectories

The language of spectroscopy readily supplies nouns: a line, a source, a wavelength, a redshift. The present inquiry is concerned with how those nouns become available, and must not place a completed *source* at the beginning of its own construction. For an opening case, take one finite registration trajectory associated with a laboratory reference procedure and another associated with an astronomical observing procedure. Their carriers and histories differ. A rule of feature identification and calibration permits parts of the trajectories to be compared, and an accepted correspondence makes it possible to write two wavelength values with compatible units and reference conventions.

Within the classical comparison the familiar parameter is

$$z = \frac{\lambda_{\text{obs}} - \lambda_{\text{ref}}}{\lambda_{\text{ref}}}. \quad (11.1)$$

The expression is a precise rule on supplied values. The words *observed* and *reference* here name positions within reports, not ready-made objects travelling intact through an unobserved interval. In an empirical use, each λ has a registration and calibration history, an uncertainty, a feature-identification decision and an admissibility condition. An astronomy report may make these more or less accessible to another reader. The compact z does not by itself enumerate them.

11.2 Distinctions that must be named

A line can change its reported position, become wider under a fitting convention, or change its separation from neighbouring features. These are not one undifferentiated *expansion*. A report should say which distinction is being compared before it asks how to interpret it. Its earlier records include detector states, calibration exposures, extraction and rejection decisions, the resolution of the apparatus and the model used to fit a feature. Existing reference resources distinguish observed and derived line positions, attach uncertainties, and retain choices such as air or vacuum wavelength conventions [5]. Survey reduction documentation describes the movement from raw and calibration exposures through masks, noise estimates, co-added spectra and fitted redshifts [6]. These examples show that

a finite-report audit can be carried out against actual documented stages, rather than inventing a generic pipeline that no observer uses.

The meta-model question is how a difference becomes distinguishable, stabilised and reportable across these two trajectories. We can inspect what is present in the full report, what is compressed into z , and which later inference would be insufficiently supported if it relied on z alone. No assumption that every spectral divergence is manufactured by notation follows from this audit. Equally, the conventional interpretation of a parameter is not licensed simply by displaying its final ratio. Its bridge is a separate, inspectable part of the scientific account.

11.3 A later physical programme

The spectral case belongs at the end of this volume because it places the arithmetic account under the pressure of a genuine scientific comparison. It should not become a claim that representation theory already supplies a quantitative alternative to existing models of light or cosmology. That further claim would require an explicit rule connecting finite registered trajectories to a family of predicted spectral comparisons, including nearby cases, varied instruments and independent checks. It would have to say what part of the recorded pattern is being accounted for and what new report could discriminate its proposed bridge from others.

It is possible to suspend a wave account while constructing the finite comparison and still acknowledge that real instruments embody earlier optical commitments. Bracketing a prior interpretation is a methodological act within a report, not a claim that the present inscription arrived without a history. The value of this chapter is to give the future experiment a clear question: which distinctions survive when the familiar spectral nouns are unfolded, and what new registration is needed to decide between two admitted continuations?

Chapter 12

Audit rules and the handover to objects

12.1 An audit of a claim

The preceding chapters suggest a repeatable method. Begin with a claim whose proposed use is known, and identify the finite report that presently supports it. Follow the particular FSTs by which its relevant registrations were formed, transformed and connected. Mark the points where a value, noun, fitted feature or mathematical result became a portable bead. Ask what the compression retained, what was explicitly moved to another column, what remains accessible in a companion record, and which distinction never entered an available registration. Then identify the exact next action or bridge the claim asks the bead to support.

If two exhibited ledgers give the same reported value and support different admissibility judgements for that action, record an insufficiency witness. If an omitted record is required to complete the claim, open a debt entry that names what is owed and where it might be sought. A later reconstruction may extend the report, narrow uncertainty or close that particular debt. It will be a new finite event and should be written as such. The audit does not demand an imaginary completed ledger of every interaction. It demands clarity about the particular closure being claimed now.

12.2 The limits of the present formalism

No general scalar measure of representational debt has been derived here. Nor have we shown that a difference between histories must alter every future measured value. The insufficiency witness is local and conditional. Even when a missing distinction is found, its relevance to a physical theory requires another bridge. These limits are part of the provisional construction. If later work supplies a metric, a theorem of sufficiency over a bounded domain or an instrumentally discriminating alternative model, it must identify the new registrations and conditions that support it.

The use of classical symbols in these chapters is intentional. A finite account can study formal equality without treating every equality as an empirical identity. The report can use sums and ratios as internal tools while documenting the conditions of their application. Part III may construct new mathematical objects and operators from explicitly finite commitments. This volume prepares that work by specifying why an object must carry

or refer to more than its extracted value when its permitted continuations depend on the enabling chain.

12.3 The object reopened

The separate outline for a monograph on mathematical objects asks what a mathematical object becomes when it is understood as a recurrently accessible stabilisation within a symbolic lattice. The work here adds a question to that definition: which finite continuations does the particular registration license, and where are the limits of its reconstruction recorded? A measured number, finite Summa or embedded transform object could be named only after its construction and admissibility have been made inspectable. Part II therefore ends at the doorway of objecthood rather than attempting to resolve all of its mathematics in one volume.

Conclusion: A correct answer and an insufficient account

The other half never needed to contradict the equation. The arithmetic $1 - \frac{1}{2} = \frac{1}{2}$ remains correct under its rules. An account of a particular taking asks a further question about disposition, history and the registered whole. Likewise, $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$ remains a correct value relation, while two finite bead ledgers yielding that value can diverge in the next admissible cut. The point of the examples is not that mathematical symbols are empty. It is that they are powerful compressions, and a compression has a range of subsequent uses under a licence.

A finite report is a document in which such licences, compressions and histories can be brought together. It is neither a transparent window onto a ready-made thing nor a mere collection of disconnected marks. Its coherence is built by following FSTs, admitting correspondences and declaring which questions the assembled account can answer. A reported value can be removed from the report and sent far beyond it. If a later operation depends upon a distinction left behind, the report must be reopened, an accessible ledger consulted, or a new registration made. Insufficiency names that particular need. Debt keeps its required entries visible without declaring them a hidden universal substance.

The same discipline can be carried into mathematical proof, scientific instrumentation and the comparison of spectral reports. The finite report lets the reader see where a noun or numeral came to stand for a chain, and what would be required to use it differently. That is the achievement this volume can presently claim. A theory of FSM objects and a physics of particular observations remain trajectories to be built from it, with their own finite carriers, conditions and bridges.

Appendix A

Prior explorations, September 2026

A.1 The route into the finite account

Part I developed the three-basin architecture and the axiom of finite representation. Its later chapters treated the FST, equality and symbolic compression as ways a constructed bead may stand for a longer chain. The working document *Finite Accounts and the Discarded Half*, prepared in September 2026, made subtraction its central specimen and asked for the contra-entry when a removed portion is represented by a one-column remainder [2]. The present volume takes that document as a historical bead within its own construction. Its vivid gold test is retained; its wider historical and foundational claims are to receive their own source study rather than being treated as proven by the arithmetic specimen.

The subsequent arithmetic exploration began with the difference between repeated addition and multiplying by a fraction smaller than one. The observation that multiplying fractions can be read as successive cuttings led to the recognition that the same multiplication symbol supports assembly and proportional selection under different constructive stories. Scaling provides a compact value description of both routes. The question of where the cut portion went then connected fractional multiplication with the earlier subtraction account. Three paths were distinguished: beads and their admitted finite actions, the ledger of those actions, and the classical manipulation of their reported values.

The eight-unit construction added a finite witness. Two ledgers reached one quarter, but only one supported a further half-cut under its original bead rule. This made the language of *conditional insufficiency* possible. The word *debt* was then narrowed from a general impression that calculation forgets its past to the particular missing distinction demanded by a later use. Finally, the term *finite report* was reserved for the wider formal document of coherent FSTs, rather than the extracted scalar value. This change of terminology is an instance of the volume's method: a useful earlier shorthand was unfolded and repaired when it obscured a needed distinction.

A.2 The spectral question retained as a question

A further September exploration asked whether this account could eventually alter the reading of cosmological spectral differences. Its most durable contribution to the present draft is methodological: the comparison itself is a finite symbolic trajectory, and its familiar nouns should be unfolded before they are asked to provide an explanation. A ratio formed from two reported wavelength values does not supply the full history of either registration.

Yet the absence of that history from the ratio does not, by itself, constitute a law for a spectral difference. Part II preserves the question and the route by which it arose. The physical work requires a later, explicit and discriminating bridge.

Appendix B

Audit card and notation ledger

B.1 A reusable finite card

A finite report can include an audit card without making the card a universal template. The card names the claim and its intended continuation, the finite registrations and ledgers accessible to the writer, the value-extraction rule, the displaced or excluded entries, the operative distinctions and uncertainty, and the licence for the proposed bridge. If an insufficiency witness exists, it names both ledgers and the action on which their admitted paths separate. If a debt remains, it states which record is missing and whether recovery, independent comparison or new registration is possible. The card closes with a scope sentence: what has been established within the present report, and what has been left open for later work.

B.2 Terms reserved across the volumes

The abbreviation FST continues to mean *Functional Symbolic Trajectory*. The words *finite* and *functional* should not be silently exchanged when introducing a new minimum-chain concept. A bead is a finite admitted distinction; its function can depend upon a longer history and handler. A ledger is an ordered accessible record of actions; a finite report is a coherent document of one or more FSTs; a reported value is a specific extraction. The sign $=$ may be used for exact equality within a declared formal domain, and the bar $|_C$ for conditioned comparison across a measurement bridge. The symbols α , δ , H and C retain the meanings introduced in Part I. A notation such as x° may later identify a measured-number registration, but no one glyph replaces its carrier, accessible history and declared conditions.

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